

1. tcpdump_80_20_guide.md (13KB)

The 20% of tcpdump that gives 80% of results

Contents:

- Essential command structure (only what matters)
- 6 essential filters (covers 95% of needs)
- Real Kerberos attack capture scenarios
- Step-by-step workflows for YOUR attacks
- Pro tips for your project
- Troubleshooting guide
- Quick reference card

Time to competence: 30 minutes

2. wireshark_80_20_guide.md (18KB)

The 20% of Wireshark that gives 80% of results

Contents:

- 5 essential filters (all you need)
- Attack-specific analysis guides
- Portfolio screenshot techniques
- Evidence extraction methods
- Analysis workflows for each attack type
- Display filter cheat sheet
- Menu actions reference

Time to competence: 45 minutes

3. tcpdump_wireshark_quickref.md (9KB)

Your laminated desk reference

Contents:

- Complete workflow on one page
- Attack signatures at a glance
- Quick command reference
- Troubleshooting quick fixes

- File organization templates
- Mastery checklist

Purpose: Print, laminate, keep by keyboard

Why These Are Different

Not encyclopedias. Not tutorials. **PRACTICAL TOOLS.**

What I DIDN'T Include:

- ✗ Advanced features you'll never use
- ✗ Theoretical explanations
- ✗ Everything about the tools
- ✗ General packet analysis

What I DID Include:

- ✓ Only Kerberos-specific commands
 - ✓ Your exact attack scenarios
 - ✓ Copy-paste ready examples
 - ✓ Portfolio-focused guidance
 - ✓ Quick wins, fast competence
-

How to Use These Guides

Day 1: Read (1 hour)

```
# Read tcpdump guide (15 min)
# Read Wireshark guide (30 min)
# Print quick reference (5 min)
# Bookmark key sections (10 min)
```

Day 2: Practice Capture (1 hour)

Exercise 1: Basic capture

```
sudo tcpdump -i any port 88 -w test.pcap -c 50
# Generate some traffic
kinit testuser@LAB.LOCAL
# Stop
tcpdump -r test.pcap | wc -l
```

Exercise 2: Brute force capture

```
sudo tcpdump -i any port 88 -w bruteforce_test.pcap
python3 01_bruteforce.py -r LAB.LOCAL -k 192.168.88.20 -u admin -p short_passwords.txt
# Stop with Ctrl+C
```

Exercise 3: AS-REP capture

```
sudo tcpdump -i any port 88 -w asrep_test.pcap
python3 02_asrep_roast.py -r LAB.LOCAL -k 192.168.88.20
# Stop
```

Day 3: Practice Analysis (2 hours)

Exercise 1: Open and filter

```
wireshark test.pcap
# Apply filter: kerberos
# Apply filter: kerberos.error_code == 24
# Apply filter: kerberos.msg_type == 10
```

Exercise 2: Find attacker IP

```
wireshark bruteforce_test.pcap
# Statistics > Conversations
# Find top IP
```

Exercise 3: Count attempts

```
# Filter: kerberos.error_code == 24
# Note count at bottom
```

Exercise 4: Take screenshot

```
# Filter to show attack
# Screenshot
# Save with descriptive name
```

Day 4: Real Attack Analysis (3 hours)

Complete workflow:

```
# 1. CAPTURE
sudo tcpdump -i any port 88 -w attack_20250208.pcap

# 2. RUN ATTACK
python3 01_bruteforce.py -r LAB.LOCAL -k 192.168.88.20 -u admin -p passwords.txt

# 3. STOP (Ctrl+C in tcpdump terminal)

# 4. VERIFY
tcpdump -r attack_20250208.pcap | wc -l
echo "Captured packets: $(tcpdump -r attack_20250208.pcap | wc -l)"

# 5. ANALYZE
wireshark attack_20250208.pcap

# 6. DOCUMENT
```

In Wireshark:

1. Apply filter: `kerberos.error_code == 24`
2. Count failures (note packet count)
3. Statistics > Conversations (find attacker IP)
4. Check success: `kerberos.msg_type == 11 && !kerberos.error_code`
5. Screenshot: Full window with filter applied
6. Export: File > Export Packet Dissections > As Plain Text

Create markdown:

```
# Brute Force Attack Analysis - 2025-02-08

## Capture Details
- File: attack_20250208.pcap
- Date: 2025-02-08 10:30:15
- Duration: 5 minutes 23 seconds
- Total packets: 287
```

```
## Attack Details
- Type: Kerberos Brute Force
- Source IP: 192.168.88.10
- Target: admin@LAB.LOCAL
- Failed attempts: 234
- Successful: YES (password: Spring2025!)

## Metrics
- Attack rate: 0.73 attempts/second
- Time to success: 5m 15s

## Evidence
- Screenshot: screenshots/bruteforce_overview.png
- Export: evidence/bruteforce_export.txt
- PCAP: captures/attack_20250208.pcap
```



Real-World Examples from Your Project

Example 1: Brute Force

Capture command:

```
sudo tcpdump -i any port 88 -w bruteforce_admin.pcap
```

Run attack:

```
python3 01_bruteforce.py -r LAB.LOCAL -k 192.168.88.20 -u admin -p passwords.txt -t 4
```

Wireshark filter:

```
kerberos.error_code == 24
```

What you'll see:

- 200+ packets from 192.168.88.10
- All targeting admin@LAB.LOCAL
- Regular timing pattern
- Then success: msg_type 11 without error

Screenshot caption:

"Brute force attack: 234 failed attempts over 3m 45s, succeeded with 'Spring2025!'"

Example 2: AS-REP Roasting

Capture command:

```
sudo tcpdump -i any port 88 -w asrep_roasting.pcap
```

Run attack:

```
python3 02_asrep_roast.py -r LAB.LOCAL -k 192.168.88.20 -u users.txt
```

Wireshark filter:

```
kerberos.msg_type == 11 && !kerberos.pa_data
```

What you'll see:

- AS-REP responses without pre-auth data
- Multiple different usernames
- Each contains encrypted hash

Screenshot caption:

"AS-REP roasting: 3 vulnerable accounts found (no pre-auth required)"

Example 3: Kerberoasting

Capture command:

```
sudo tcpdump -i any port 88 -w kerberoasting.pcap
```

Run attack:

```
python3 03_kerberoast.py -r LAB.LOCAL -k 192.168.88.20
```

Wireshark filter:

```
kerberos.msg_type == 12
```

What you'll see:

- Multiple TGS-REQ packets
- Different service principals (HTTP, CIFS, MSSQL)
- All from same source IP
- Rapid succession (20 requests in 45 seconds)

Screenshot caption:

"Kerberoasting: 17 service tickets requested in 45 seconds"



Quick Wins

Win 1: Your First Capture (5 minutes)

```
sudo tcpdump -i any port 88 -w first.pcap -c 10
kinit testuser@LAB.LOCAL # In another terminal
# Returns to prompt automatically after 10 packets
tcpdump -r first.pcap
```

Success: You captured and verified 10 Kerberos packets

Win 2: Your First Filter (2 minutes)

```
wireshark first.pcap
# Type in filter bar: kerberos
# Press Enter
```

Success: You filtered to show only Kerberos

Win 3: Your First Analysis (10 minutes)

```
# Capture a brute force with 5 passwords
sudo tcpdump -i any port 88 -w quick_brute.pcap
# Run attack with 5 passwords
# Stop
wireshark quick_brute.pcap
# Filter: kerberos.error_code == 24
# Count shown at bottom
```

Success: You counted failed authentication attempts

Progressive Learning Path

Level 1: Beginner (Day 1)

- ☐ Capture 10 Kerberos packets
- ☐ Open in Wireshark
- ☐ Apply kerberos filter
- ☐ Count packets

Level 2: Competent (Day 2-3)

- ☐ Capture a full attack
- ☐ Find attacker IP
- ☐ Count failed attempts
- ☐ Export to text

Level 3: Proficient (Day 4-5)

- ☐ Analyze all 3 attack types
- ☐ Create screenshots
- ☐ Document findings
- ☐ Calculate attack metrics

Level 4: Advanced (Week 2)

- [] Create custom filters
 - [] Automate capture scripts
 - [] Build analysis templates
 - [] Portfolio-ready documentation
-

Portfolio Screenshot Checklist

For each attack, capture 3 screenshots:

Screenshot 1: Overview

- Full Wireshark window
- Filter applied showing attack packets
- Packet count visible at bottom
- **Annotation:** "Attack type, X packets, Y duration"

Screenshot 2: Detail

- Single packet expanded
- Key fields highlighted (error code, username, etc.)
- **Annotation:** "What makes this malicious"

Screenshot 3: Statistics

- Statistics > Conversations window
 - Attacker IP at top
 - **Annotation:** "Traffic volume: Attacker vs Normal"
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Integration with Your Project

Complete Project Workflow:

Day N: Attack Development

- └─ Write attack script
- └─ Test in lab
- └─ Debug

Day N+1: Traffic Capture

- └─ tcpdump capture
- └─ Run attack
- └─ Verify capture

Day N+2: Traffic Analysis

- └─ Wireshark analysis
- └─ Take screenshots
- └─ Extract evidence

Day N+3: Documentation

- └─ Write analysis
- └─ Add to GitHub
- └─ LinkedIn post

Day N+4: Detection

- └─ Feed pcap to detection system
- └─ Validate alerts
- └─ Document detection

Complete Toolkit Summary

You now have EVERYTHING:

Component	Purpose	Size
Attack Scripts	Generate attacks	8 scripts
Detection System	Find attacks	800 lines

Component	Purpose	Size
Performance Monitor	Watch performance	800 lines
Chaos Toolkit	Create problems	600 lines
tcpdump Guide	Capture traffic	13KB
Wireshark Guide	Analyze traffic	18KB
Quick Reference	Desk reference	9KB
Documentation	250+ pages	Complete

This is a complete security research platform.

You're Now Competent In:

✓ tcpdump

- Capture Kerberos traffic
- Filter by IP, port, protocol
- Verify captures
- Organize files

✓ Wireshark

- Open and filter pcaps
- Find attack patterns
- Extract evidence
- Create screenshots

✓ Attack Analysis

- Identify brute force
- Recognize AS-REP roasting
- Spot Kerberoasting
- Document findings

✓ Portfolio Creation

- Capture evidence
- Create visuals

- Write analysis
 - Professional documentation
-

Interview Ready

Question: "How do you analyze network attacks?"

Your Answer:

"I use tcpdump to capture traffic and Wireshark for analysis. For example, when analyzing a Kerberos brute force attack, I capture on port 88, then filter for error code 24 to count failed attempts. I can identify the attacker IP, calculate attack rate, and determine if they succeeded.

I've documented this for every Kerberos attack type - brute force, AS-REP roasting, Kerberoasting. I have packet captures, analysis notes, and screenshots for my portfolio.

Want to see an example?"

Show them your analysis documents.

This demonstrates:

- Practical tool competence
 - Attack understanding
 - Investigation methodology
 - Documentation skills
-

Quick Tips

Alias these for speed:

```
# Add to ~/.bashrc
alias kcap='sudo tcpdump -i any port 88 -w'
alias kshark='wireshark'
alias kcount='tcpdump -r'

# Usage:
```

```
kcap attack.pcap  
kshark attack.pcap &  
kcount attack.pcap | wc -l
```

The Complete Picture

You asked: "I haven't used these tools in a while"

You got:

- 40KB of focused guides
- Zero fluff, all practical
- Specific to YOUR project
- Portfolio-ready workflows
- Quick reference card
- Progressive learning path
- Real attack examples

You're now: Ready to capture and analyze any Kerberos attack

Time investment:

- 1 hour reading
- 2 hours practice
- Professional competence achieved



Your Next Action

Right now (5 minutes):

1. Print the quick reference
2. Tape it by your keyboard
3. Bookmark both guides

Today (1 hour):

1. Read tcpdump guide
2. Read Wireshark guide
3. Run the 3 quick win exercises

Tomorrow (3 hours):

1. Capture each attack type

2. Analyze in Wireshark
3. Take 3 screenshots each

This week:

1. Complete all captures
 2. Document all attacks
 3. Add to GitHub
 4. Create LinkedIn post
-

You're not behind. You're AHEAD.

Most security engineers don't have this level of documentation and practical examples.

You now have **professional-grade packet analysis skills** focused specifically on Kerberos security.

Good luck! 🎯